

DEEP LEARNING WITH CNNs & TRANSFER LEARNING

In this assignment, you will:

1. Select a real-world image dataset of your choice.
2. Build and train a baseline CNN from scratch.
3. Apply and fine-tune at least two pre-trained transfer learning models.
4. Compare performance, analyze results, and write a technical report.

Dataset Requirements

You may choose any publicly available image dataset that satisfies the following:

1. Task type: Image classification (binary or multi-class).
2. Size: At least 2,000 images in total (can be more).
3. Classes: Minimum 3 classes (preferably 5+).
4. Split suggestion: ~70% train, 15% validation, 15% test.
5. Examples (you may use, but are not limited to):
 - Medical images (e.g., X-ray, OCT, skin lesion, etc.)
 - Natural images (animals, objects, scenes)
 - Remote sensing / satellite imagery
 - Traffic signs, faces, etc.

Implementation Details:

1. Baseline CNN from Scratch
2. Transfer Learning with Pre-Trained Models (Minimum 3)
 - a. ResNet (e.g., ResNet50)
 - b. DenseNet (e.g., DenseNet121)
 - c. EfficientNet (e.g., B0/B1)
 - d. MobileNetV2 / MobileNetV3
 - e. InceptionV3 / Xception
 - f. VGG16 / VGG19
 - g. (Optional/advanced) Vision Transformer (ViT, Swin, etc., if you're comfortable)
3. Evaluations:

- a. Training and validation curves.
 - b. Test accuracy
 - c. Precision, recall, F1-score (per class or macro).
 - d. Top-k accuracy (for multi-class).
 - e. Confusion matrices for each model.
4. Efficiency analysis
- a. Number of parameters,
 - b. Training time per epoch (approximation),
 - c. Inference time per image (optional).
5. Compare: Baseline CNN vs Transfer Learning models
6. (Optional but Highly Recommended) Explainability
- a. Apply at least one basic explainability/XAI method to your best model:
 - i. Grad-CAM / Grad-CAM++
 - ii. Saliency maps
 - b. Show visualizations for 4–6 correctly classified and misclassified images and comment:
 - i. Are the activations focusing on relevant regions of the image?
 - ii. Does this increase your trust in the model predictions?

REPORT:

1. Title & Abstract
 - a. Short, meaningful title.
 - b. 150–200 word abstract summarizing dataset, methods, and key findings.
2. Introduction
 - a. Motivation and problem statement.
 - b. Short description of your dataset and task.
 - c. Contribution: What exactly did you implement/compare?
3. Methodology
 - a. Dataset details and preprocessing.
 - b. Baseline CNN architecture.
 - c. Transfer learning models and how you fine-tuned them.

- d. Training details and hyperparameters.
- 4. Experiments & Results
 - a. Training/validation curves.
 - b. Tables of metrics.
 - c. Confusion matrices.
 - d. Misclassified examples.
 - e. (Optional) XAI visualizations.
- 5. Discussion
 - a. Comparative analysis of models.
 - b. Observations about overfitting, generalization, class imbalance, etc.
- 6. Conclusion & Future Work
 - a. Summary of what you achieved.
 - b. Limitations and possible improvements (e.g., larger dataset, different architectures, better augmentation, advanced regularization).
- 7. References
 - a. Cite at least 5 relevant papers (original CNN/ResNet papers, dataset papers, etc.).